

Variables

Problem 1: Variables of States

Suppose that the fifty states are the observational units. For each item, write **yes, it is a variable** or **no, it is not a variable**.

For each variable, report or make a reasonable guess for its value in your home state. For each item that is not a variable, briefly explain why.

- a) The number of states that have a female governor
- b) The percentage of the state's residents who are over 65 years of age
- c) The highest speed limit in the state
- d) Whether the state's name consists of one word
- e) The average income of an adult resident of the state
- f) The number of births per 1,000 residents of the state in 1999
- g) The number of states that were settled before 1865
- h) The average number of Congressional representatives per state
- i) The number of Congressional representatives from the state

Problem 2: Types of Variables

The observational units are 22 Broadway productions. Classify each variable from the supplied dataset as **quantitative** or **categorical**. If a categorical variable has exactly two possible values, also identify it as **binary**.

- a) Production
- b) Type
- c) Gross
- d) Attendance
- e) Capacity
- f) Top price

Dataset: Broadway Productions

Production	Type	Gross	Attendance	Capacity	Top price
Annie Get Your Gun	Musical	572885	9892	12761	75
Cabaret	Musical	466670	6647	7360	90
Cats	Musical	350368	7200	11856	70
Chicago	Musical	536852	9664	11680	80
Death of a Salesman	Play	351082	6531	7483	75
Epic Proportions	Play	80379	2564	4776	65
Footloose	Musical	271022	6974	10784	75
Fosse	Musical	566644	8118	9128	80
It Ain't Nothin' But The Blues	Musical	88014	2366	5500	75
Jekyll & Hyde	Musical	283674	5613	8512	80
Kat and the Kings	Musical	108091	3218	8672	75
Les Miserables	Musical	375318	7677	11336	80
Miss Saigon	Musical	395522	8558	13945	80
Ragtime	Musical	420902	8913	14495	75
Rent	Musical	333248	6219	9448	80
Side Man	Play	106437	2894	6440	60
Smokey Joe's Cafe	Musical	168276	3547	9697	75
The Lion King	Musical	880717	14199	14032	80
The Phantom of the Opera	Musical	601218	9974	12856	80
The Scarlet Pimpernel	Musical	125597	2284	5464	80
The Weir	Play	120605	3450	7576	60
Voices in the Dark	Play	114607	3848	8688	60